# ON AN INTEGRAL NORMAL SCHEME THE PUSHFORWARD OF A CONSTANT SHEAF ALONG THE GENERIC POINT OR A NONEMPTY OPEN IS AGAIN CONSTANT

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the proof is written out in full for both parts: the normality of the strict henselization and the resulting irreducibility for the first, and the injectivity from the density of the generic fibre for the second.

*Editorial note.* The label lemma-push-constant-sheaf-from-open names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

*Allowed standard prerequisites (editorial).*

- Strict henselisations of a normal local ring are normal domains. More on Algebra, the lemma stating that a strict henselization of a normal local ring is a normal domain. Supplies the irreducibility of the base change Spec(K) x_X Spec(O^sh_{X,x}).
- Stalks of a pushforward as cohomology of the fibre. Same chapter, the theorem describing the stalk of a pushforward as the cohomology of the corresponding fibre product. Supplies the identification of the stalk of g_*underline E with E.
- Density of the generic fibre over an etale neighbourhood. Standard property of an integral scheme, used in the second part: for V etale over X the scheme Spec(K) x_X V is dense in U x_X V for a nonempty open U. Supplies the injectivity of the map j_*underline E -> g_*underline E.

*Source reference.* Original lemma 06DI: sources/stacks-chow/tags.txt, source lines 8182–8182. Prerequisite pointer invoked at proof line(s) 12715; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 03Q9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8015–8015. Prerequisite pointer invoked at proof line(s) 12720; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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*Date*: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $X$ be an integral normal scheme with function field $K$. Let $E$ be a set.*

(1) *Let $g : \mathrm{Spec}(K) \to X$ be the inclusion of the generic point. Then $g_*\underline{E} = \underline{E}$.*
(2) *Let $j : U \to X$ be the inclusion of a nonempty open. Then $j_*\underline{E} = \underline{E}$.*

*Proof.* Proof of (1). Let $x \in X$ be a point. Let $\mathcal{O}^{sh}_{X,\overline{x}}$ be a strict henselization of $\mathcal{O}_{X,x}$. By More on Algebra, Lemma 06DI we see that $\mathcal{O}^{sh}_{X,\overline{x}}$ is a normal domain. Hence $\mathrm{Spec}(K) \times_X \mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}})$ is irreducible. It follows that the stalk $(g_*\underline{E})_{\overline{x}}$ is equal to $E$, see Theorem 03Q9.

Proof of (2). Since $g$ factors through $j$ there is a map $j_*\underline{E} \to g_*\underline{E}$. This map is injective because for every scheme $V$ étale over $X$ the set $\mathrm{Spec}(K) \times_X V$ is dense in $U \times_X V$. On the other hand, we have a map $\underline{E} \to j_*\underline{E}$ and we conclude. □